

CONCEPCION, Paraguay

WMO: 861340

Lat: 23.442S

Lon: 57.427W

Elev: 77

StdP: 100.40

Time Zone: -4.00 (PRY)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.1	8.3	1.5	4.2	11.6	3.4	4.9	11.8	10.4	26.4	9.4	25.7	1.2	180	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	37.8	23.9	36.8	24.1	35.8	24.2	27.1	32.7	26.7	32.3	26.4	32.1	5.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.1	29.8	25.2	20.5	29.5	24.7	19.9	29.2	86.0	32.6	84.1	32.2	82.8	33.4	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.4	7.7	2.4	39.7	2.3	1.2	0.8	40.5	-0.5	41.2	-1.8	41.9	-3.5	42.7
			1.9	27.9	2.1	0.3	0.3	28.1	-0.9	28.3	-2.1	28.5	-3.7	28.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	28.4	28.0	27.0	24.6	20.9	19.6	19.8	21.6	23.3	25.5	26.1	27.9
	DBStd	5.07	1.87	2.30	2.58	3.64	3.97	5.04	5.35	5.66	5.29	3.87	3.03	2.53
	HDD10.0	3	0	0	0	0	0	1	2	1	0	0	0	0
	HDD18.3	177	0	0	0	5	20	48	52	35	16	1	0	0
	CDD10.0	5259	572	505	527	439	339	290	304	361	401	481	484	557
	CDD18.3	2391	313	272	268	193	101	87	96	137	167	224	234	298
	CDH23.3	25588	3663	3034	2806	1765	766	637	843	1532	1987	2492	2597	3467
(14) Wind	WSAvg	3.1	2.7	2.7	2.4	2.6	2.7	3.1	3.5	3.8	3.9	3.7	3.3	3.1
	PrecAvg	1422	168	151	139	138	117	72	38	27	75	145	195	158
	PrecMax	2135	427	478	301	332	231	277	131	110	220	258	477	385
	PrecMin	925	43	24	20	3	20	2	0	0	4	23	61	41
	PrecStd	267	105	90	69	92	64	63	31	28	54	58	109	84
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	38.1	37.7	35.7	33.2	32.3	33.4	37.0	38.7	39.6	38.3	37.7
		MCWB	24.3	25.0	24.5	25.3	23.6	22.3	20.9	21.1	22.6	23.8	24.9	24.8
	2%	DB	36.4	36.2	36.1	34.4	31.6	30.6	31.8	35.2	37.2	36.8	36.4	36.5
		MCWB	24.8	24.7	24.4	25.0	23.5	22.0	20.9	21.2	22.4	24.2	24.6	25.2
	5%	DB	35.1	35.0	34.7	33.0	30.0	29.3	30.3	33.4	35.5	35.2	34.8	35.2
		MCWB	24.9	25.2	24.8	24.7	23.1	21.9	20.8	20.7	22.2	24.0	24.8	25.3
	10%	DB	33.9	33.5	33.1	31.5	28.1	27.6	28.6	31.4	33.2	33.4	33.1	33.9
		MCWB	25.2	25.2	24.8	24.3	22.4	21.5	20.5	20.6	21.8	23.9	24.3	25.5
	0.4%	WB	27.5	27.6	27.2	26.7	25.6	24.7	23.7	24.0	25.5	26.7	27.4	27.5
		MCDB	32.6	32.9	32.4	32.7	30.3	29.1	29.3	31.1	33.2	34.0	33.5	33.2
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.8	26.8	26.4	26.1	24.7	23.5	22.8	22.9	24.3	26.0	26.4	26.7
		MCDB	31.9	32.2	32.2	31.9	29.2	28.2	28.9	30.5	32.8	33.2	32.9	32.5
	5%	WB	26.5	26.4	25.8	25.5	23.8	22.6	21.8	22.1	23.4	25.3	25.6	26.4
		MCDB	31.7	31.9	31.7	31.0	28.1	27.2	27.8	30.1	31.8	32.1	32.1	32.3
	10%	WB	26.1	25.9	25.4	24.9	22.9	21.7	20.9	21.2	22.5	24.5	25.1	26.0
		MCDB	31.4	31.2	31.2	30.0	27.1	26.3	26.8	29.3	30.6	31.2	31.3	31.8
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	9.7	10.3	9.8	9.2	9.5	11.0	12.7	12.6	11.0	10.9	9.7
		MCDBR	12.2	11.9	12.9	11.8	11.6	11.4	13.3	14.8	15.3	13.5	13.2	11.9
		MCWBR	3.6	3.5	4.0	4.4	5.2	5.1	5.5	5.2	5.2	4.6	4.5	3.8
	5% WB	MCDBR	9.4	9.4	10.3	10.3	9.7	9.8	11.5	12.8	12.8	11.5	10.4	9.6
		MCWBR	3.8	3.8	4.1	4.6	5.3	5.3	6.1	6.0	5.3	4.9	4.4	4.0
(40) Clear-Sky Solar Irradiance	taub		0.416	0.414	0.399	0.400	0.379	0.401	0.402	0.515	0.609	0.525	0.428	0.419
	taud		2.405	2.414	2.443	2.417	2.439	2.370	2.347	1.979	1.764	2.041	2.327	2.378
	Ebn at Noon		930	917	904	858	835	788	801	735	709	810	912	928
	Edh at Noon		127	124	115	109	98	101	106	166	223	177	137	131
(44) All-Sky Solar Radiation	RadAvg		6.68	6.01	5.56	4.62	3.42	2.97	3.30	4.02	4.68	5.42	6.27	6.62
	RadStd		0.45	0.46	0.34	0.42	0.49	0.33	0.37	0.28	0.47	0.51	0.39	0.41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page